Uniform Square-Root-Mass Control of Wilson-Dirac Boundary Covariances

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Abstract

For the one-dimensional two-orbital Wilson–Dirac occupied-band projections, we establish a mass-uniform trace-class estimate for the difference of the cross-boundary blocks of the infinite-chain projections compressed to a finite interval. If the interval contains L cells, the mass is $m = \mu/L$, and $\log(1+m) \le 1/4$, then the boundary trace norm is at most $C\sqrt{\mu}$, for a universal constant C. Consequently, the quasifree interval entropies differ by at most $2C\sqrt{\mu}$. The proof uses a Fourier–Laplace/Hankel decomposition, retaining the finite-length square-root weight throughout the branch-point estimates. Combining this with the fixed-μ Hilbert–Schmidt continuum theorem shows that the exponent $1/2$ cannot be increased for a uniform estimate on the boundary trace norm. The entropy-difference limit is not determined by this bound.

1. Setting and theorem

Let P_m be the orthogonal Fourier-multiplier projection on $\ell^2(\mathbb{Z};\mathbb{C}^2)$ associated with the negative-energy band of

$$h_m(k) = \sin k\,\sigma_x + (1+m-\cos k)\sigma_z, \qquad P_m(k) = \tfrac{1}{2}(I - h_m(k)/E_m(k)),$$
$$E_m(k) = \sqrt{m^2 + 4(1+m)\sin^2(k/2)}.$$

The massless projector P_0 is defined almost everywhere. Let χ_L project onto any L consecutive two-orbital cells, and set

$$m = \mu/L > 0, \quad \kappa = \log(1+m), \quad Z_{L,\mu} = \chi_L(P_m - P_0)(I - \chi_L).$$

Theorem 1. There is a universal finite constant C such that, for every $L \ge 1$ and $\mu > 0$ satisfying $0 < \kappa \le 1/4$,

$$\boxed{\|Z_{L,\mu}\|_{S_1} \le C\sqrt{L\kappa} \le C\sqrt{\mu}.} \tag{1}$$

For the compressed covariance matrices $A_L = \chi_L P_m \chi_L$ and $B_L = \chi_L P_0 \chi_L$, on the interval subspace,

$$\boxed{|S(A_L) - S(B_L)| \le 2C\sqrt{\mu}.} \tag{2}$$

For every fixed $\mu > 0$, the condition on κ holds for all sufficiently large L. The constant does not depend on L or μ in that range.

2. Exact Fourier–Laplace representation

Write $a = e^{\kappa} = 1 + m$, and define, for $t > 0$,

$$h_\kappa(t) = \mathbf{1}_{t>\kappa}\frac{e^{-\kappa/2}}{\sqrt{2(\cosh t - \cosh\kappa)}}, \qquad h_0(t) = \frac{1}{2\sinh(t/2)},$$

$$\rho_x(t) = \sinh t\,[h_\kappa(t) - h_0(t)], \qquad \rho_z(t) = (e^{\kappa} - \cosh t)h_\kappa(t) - (1-\cosh t)h_0(t).$$

For $n \geq 1$, the positive-distance Fourier coefficient of $P_m - P_0$ is

$$T_n = -\frac{1}{2\pi} \int_0^\infty e^{-nt} [\rho_z(t)\sigma_z + i\rho_x(t)\sigma_x] dt, \qquad T_{-n} = T_n^*. \tag{3}$$

This follows by writing $E_m(k) = e^{\kappa/2}\sqrt{2(\cosh\kappa - \cos k)}$, inserting the Poisson kernel, and taking the cosine/sine Fourier coefficients. In particular the normalized cosine representation is

$$\frac{1}{\pi}\int_0^\pi \frac{\cos(nk)}{\sqrt{2(\cosh\kappa - \cos k)}} dk = \frac{1}{\pi}\int_\kappa^\infty \frac{e^{-nt}}{\sqrt{2(\cosh t - \cosh\kappa)}} dt.$$

All subsequent integrals involve the combination (3), in which the singular massless terms cancel; we do not assign a finite value to the individually divergent massless inverse-energy Fourier transform.

Derivation of the normalized transform. For $\kappa > 0$ set $g_\kappa(k) = [2(\cosh\kappa - \cos k)]^{-1/2}$. The Poisson kernel identity

$$\sum_{n\in\mathbb{Z}} e^{-|n|t} e^{ink} = \frac{\sinh t}{\cosh t - \cos k}$$

can be integrated against $[2(\cosh t - \cosh\kappa)]^{-1/2}\, dt$ over $t > \kappa$. The substitution $u = \sqrt{2(\cosh t - \cosh\kappa)}$ reduces the resulting integral to

$$\int_0^\infty \frac{du}{u^2/2 + \cosh\kappa - \cos k} = \pi g_\kappa(k).$$

Comparison of Fourier coefficients proves the displayed cosine transform. Multiplication by $1 + m - \cos k$ or $\sin k$ gives the z and x coefficients in (3). The limit $\kappa \downarrow 0$ is taken in these combinations, where the singular terms cancel.

The right and left exterior half-lines each give a matrix-valued Hankel block. Rank-one factorization of $e^{-t(i+j+1)}$, for $0 \leq i < L$, $j \geq 0$, has nuclear norm

$$W_L(t) = e^{-t}\frac{\sqrt{1 - e^{-2Lt}}}{1 - e^{-2t}}. \tag{4}$$

Taking the Schatten norm of the 2×2 orbital factor in (3), and accounting for both boundaries, yields the rigorous Bochner-integral estimate

$$\|Z_{L,\mu}\|_{S_1} \leq \frac{2}{\pi}\int_0^\infty W_L(t)(|\rho_x(t)| + |\rho_z(t)|) dt. \tag{5}$$

3. Strengthened finite-length kernel bound

The essential improvement is to keep the finite-L numerator of (4) instead of prematurely replacing it by one. Because $1 - e^{-2Lt} \leq 2Lt$ and $1 - e^{-2t} \geq (1 - e^{-2})t$ when $0 < t \leq 1$, there is a universal C_0 such that

$$\boxed{W_L(t) \leq C_0\sqrt{L/t} \quad (0 < t \leq 1), \qquad W_L(t) \leq C_0 e^{-t} \quad (t \geq 1).} \tag{6}$$

The bound need not be optimal for large t; its small-t behavior is what retains the $\sqrt{L\kappa}$ scale.

4. Four-region proof

Assume $0 < \kappa \leq 1/4$. All constants in this section are universal.

Region I: $0 < t < \kappa$**.** The massive density vanishes: $h_\kappa = 0$. Exactly,

$$\rho_x = -\cosh(t/2), \qquad \rho_z = \sinh(t/2).$$

Thus $|\rho_x| + |\rho_z| \le C_1$, and (6) gives

$$\int_0^\kappa W_L(|\rho_x| + |\rho_z|)dt \le C_2\sqrt{L}\int_0^\kappa t^{-1/2}dt \le C_3\sqrt{L\kappa}. \tag{7}$$

Region II: $\kappa < t < 2\kappa$**.** Elementary bounds on $\cosh t - \cosh \kappa$, the massive prefactor, and the massless density show

$$|\rho_x| + |\rho_z| \le C_4\left(1 + \frac{t+\kappa}{\sqrt{t^2-\kappa^2}}\right).$$

Set $t = \kappa u$ in (6). Then

$$\int_\kappa^{2\kappa} W_L(|\rho_x| + |\rho_z|)dt \le C_5\sqrt{L\kappa}\int_1^2 u^{-1/2}\left(1 + \frac{u+1}{\sqrt{u^2-1}}\right)du. \tag{8}$$

The final integral is finite: its singularity at $u = 1$ is only of order $(u-1)^{-1/2}$. Hence (8) is $O(\sqrt{L\kappa})$.

Region III: $2\kappa < t < 1$**.** Since $t \ge 2\kappa$, the denominator of h_κ is uniformly comparable with t; the exponential prefactor changes by $O(\kappa)$ and the denominator by $O(\kappa^2)$. Consequently,

$$|h_\kappa - h_0| \le C_6(\kappa/t + \kappa^2/t^3),$$

$$|\rho_x| \le C_7(\kappa + \kappa^2/t^2), \qquad |\rho_z| \le C_7\kappa/t.$$

Using (6), the region-III integral is bounded by

$$C_8\sqrt{L}\left[\kappa\int_{2\kappa}^1 t^{-1/2}dt + \kappa^2\int_{2\kappa}^1 t^{-5/2}dt + \kappa\int_{2\kappa}^1 t^{-3/2}dt\right] \le C_9\sqrt{L\kappa}. \tag{9}$$

All exponents have been kept explicit; the potentially difficult lower endpoint contributes only $O(\sqrt{L\kappa})$.

Region IV: $t \ge 1$**.** The inverse-energy denominators are separated from their singularities and

$$|\rho_x| + |\rho_z| \le C_{10}\kappa e^{t/2}.$$

With $W_L \le C_0 e^{-t}$, the final integral is $O(\kappa)$, which is $O(\sqrt{L\kappa})$ for $L \ge 1$ and $\kappa \le 1$.

Adding the estimates (7)–(9) and the fourth region in (5) proves the first assertion of Theorem 1. Finally $L\log(1+\mu/L) \le \mu$, proving the second inequality in (1).

5. Entropy continuity

For any infinite-space orthogonal projector P, let $X = \chi_L P(I-\chi_L)$ and $C = \chi_L P\chi_L$ on the interval. Then $XX^* = C(I-C)$. The singular-value entropy function

$$F(s) = h\left(\frac{1+\sqrt{1-4s^2}}{2}\right), \quad 0 \le s \le \tfrac{1}{2},$$

has derivative

$$F'(s) = \frac{4s}{\sqrt{1-4s^2}}\,\operatorname{artanh}\sqrt{1-4s^2}, \quad 0 \le F'(s) \le 2.$$

Mirsky's singular-value inequality and the scalar Lipschitz bound imply

$$|S(A_L) - S(B_L)| \le 2\|\chi_L(P_m - P_0)(I-\chi_L)\|_1.$$

Inserting (1) proves (2).

6. Optimality of the trace-norm exponent in the small-mass limit

The exponent in (1) cannot be raised uniformly. This conclusion also uses the separately established fixed-μ Hilbert–Schmidt continuum theorem; it does not follow from the four-region upper bound alone.

Indeed, write $\Delta_L = \chi_L(P_m - P_0)\chi_L$, $G_L = \chi_L(P_m - P_0)^2\chi_L$. Exact positive compression gives

$$\|Z_{L,\mu}\|_2^2 = \mathrm{Tr}(G_L - \Delta_L^2).$$

For fixed $\mu > 0$, the full-line Fourier trace and the continuum covariance-distance result in *Hilbert-Schmidt Scaling Limit for Wilson-Dirac Covariances* show

$$\lim_{L\to\infty} \|Z_{L,\mu}\|_2^2 = \Lambda(\mu) = \frac{\mu}{\pi} - H(\mu).$$

The continuum Hilbert–Schmidt scaling function is

$$H(\mu) = \frac{1}{\pi^2}\int_0^1 (1-s)\left[\mu^2 K_0(\mu s)^2 + \left(\frac{1}{s} - \mu K_1(\mu s)\right)^2\right] ds.$$

The classical small-argument Bessel expansions imply

$$K_0(z) = O(1 + |\log z|), \qquad K_1(z) - z^{-1} = O(z(1 + |\log z|)), \quad 0 < z \le 1,$$

so $H(\mu) = O(\mu^2(1 + |\log \mu|)^2) = o(\mu)$ as $\mu \downarrow 0$. Therefore

$$\lim_{\mu\downarrow 0} \frac{\Lambda(\mu)}{\mu} = \frac{1}{\pi}.$$

Since the nuclear norm dominates the Hilbert–Schmidt norm, we conclude

$$\boxed{\liminf_{\mu\downarrow 0} \liminf_{L\to\infty} \frac{\|Z_{L,\mu}\|_1}{\sqrt{\mu}} \ge \frac{1}{\sqrt{\pi}} > 0.} \tag{10}$$

Thus no estimate $\|Z_{L,\mu}\|_1 \le C\mu^{1/2+\epsilon}$, for any $\epsilon > 0$, can hold uniformly for all sufficiently large L and small μ in this sequence. **This optimality statement applies to the boundary trace norm, not to the entropy difference.**

7. Scope

The proof uses the specific Wilson–Dirac Bloch projectors, a finite interval of consecutive cells, and the mass scaling $m = \mu/L$. It does not establish uniformity for arbitrary mass sequences or other lattice Hamiltonians. It does not prove convergence of $S(A_L) - S(B_L)$ as $L \to \infty$ for a fixed positive μ. Nor does it establish a proposed finite entropy contribution, a uniform reference-modular spectral approximation, or the full relative-entropy logarithmic coefficient.

The stronger boundary convergence and entropy-difference limit are proved in *Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States*. The complete leading logarithm is established in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. These are separate arguments. Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* combines the boundary limit with modular and nonlinear convergence to obtain the complete fixed-positive-scaled-mass finite part and an $O_\mu(1/\log L)$ remainder. The bound proved in this article alone does not establish that limit.

References

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